Deep Learning Based CCTV Video Captioning for Intelligent Surveillance

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Abstract—The ability of Qwen3-VL-4B-Instruct to generate text descriptions of CCTV and urban videos has been tested in this study. Ten short videos were used as an input, and the system provided a BLEU-4 of 0.0101, a METEOR of 0.2342, and a ROUGE-L of 0.1339. The automatic summarization of videos may decrease the amount of work and time spent for inspection by enabling a user to read a summary instead of watching the video. No training and fine-tuning were done to the model, and it was used in its default settings. In the course of testing, up to 8 frames from each MP4 video were selected and resized so that the side length of the largest image dimension was equal to 768 pixels. Such prepared images served as an input for the model, and the deterministic generation of a caption in up to 180 tokens occurred. No external object detector or action recognition model was used. Each video was associated with one reference caption only. The captions obtained by the system differed significantly from the reference in length (79-141 vs. 10-18 words). The difference in length may have influenced the BLEU score because the model was forced to put a lot

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I. Introduction

CCTV cameras take a large amount of video daily. Operators do not have enough hours per day to review all of the video recorded. It can be time-consuming to go through hours of video to find anything interesting. If operators were able to see a short written description of each video, they would be able to know what happened in the video without having to watch it repeatedly.

A. Background

Cameras for video surveillance are used on roads, junctions, outdoor places, and indoors. They record continuously in these places. When a user needs a particular scene recorded by the camera, they may need to spend long hours browsing the video looking for some particular scenes

This is exhausting as well as tedious. A natural language description may prove beneficial in terms of highlighting the main content of the video in a few words. It may also be stored along with the video and later retrieved through text.

Previous works have indicated that the combination of object, action, and their context could assist the video surveillance task [1]. Likewise, individual-based descriptions would provide more insights about the visual characteristics and activities of the individual in the scene [2].

Yet another option in tackling this issue is the vision-language model, which takes visual inputs with a text prompt and generates a response using natural language. As a result of pretraining, such models possess the capability of understanding images and responding to commands. Thus, it is possible to explore whether a trained model will be capable of describing CCTV video footage without having a new feature extractor and decoder created for it.

B. Problem Statement

Most of the videos taken from CCTVs are generally in the form of video files; hence, making them harder to browse, search, and summarization. Even though there are many video-captioning models that have been designed using some general dataset, their effectiveness on the surveillance videos could differ since there is likely going to be low image resolution, occlusion, and limited movements present in the videos.

Having to train another new model for the specific task will also require data and computational resources, which can prove to be impractical for a research project of this scale.Hence, in this project, there is going to be an effort of producing descriptive captions of surveillance videos and urban videos through one vision-language model and compare the results to the captions made by humans.

Furthermore, it is difficult to generate accurate captions for surveillance videos because some important activities may only be observed in several seconds and other frames may contain repeated or ambiguous visual information.

C. Study objectives

The main objective of this work is to evaluate the performance of the pretrained Qwen3-VL-4B-Instruct model in CCTV video captioning without fine-tuning.The preprocessing steps used in the study are uniform frame sampling and image resizing. It generates a natural-language caption for each of the 10 MP4 video clips, and stores the results and relevant processing information in CSV and JSON files.

We evaluate caption quality using BLEU-4, METEOR and ROUGE-L with one reference caption per video. The caption generation time is also measured on a Tesla T4 GPU in Google Colab.In addition to reporting these results, the study also discusses the extent to which the evaluation scores can inform us about caption quality and the limits of their usefulness.

D. Acknowledgments

In this paper we provide a documented evaluation of a pretrained vision-language model for describing CCTV and urban-scene videos. The model is tested on a small number of video clips in a fixed inference setting without any further training or fine-tuning.

The evaluation includes the time taken to generate captions, the metric scores, and the individual results for each video. It also discusses the limitations of using text-overlap metrics alone to determine whether a caption correctly describes a scene.

It does not propose any new framework and does not compare the proposed model with any other model either. The framework does not aim to identify any unusual incidents nor does it make any claim about being ready for surveillance in the actual environment. The project simply presents a miniature yet replicable research of the capabilities of an existing model.

The rest of this paper proceeds as follows. In section II, previous work on the subject will be reviewed to reveal the research gap. Section III elaborates on the methodology used in this study. The fourth section discusses the data, experimental setting, and performance measures used. Section V outlines and discusses the results. Section VI focuses on limitations of this study, while Section VII concludes the paper.

II. Literature Survey

Various ways of generating captions for videos have been discussed by researchers, ranging from techniques tailored for the purpose of surveillance videos to video understanding models. But none of the literature surveyed in this part compares a pretrained vision-language model against CCTV videos through the same experimental methodology adopted in this study.

A. Captioning with Surveillance Focus

A paper presented in the Sensors journal in 2024 [1] created a specially made dataset that included image captions and danger scores. In their research, the authors used BLIP-2 to generate captions and BERT to process them and recognize hazards at various degrees of danger. Classification accuracy of 92.3% for safety, 89.8% for hazard, and 94.3% for danger was achieved.

Nonetheless, this methodology emphasizes individual images as opposed to the temporal data contained in full videos. Moreover, the authors also mentioned that more research was required to determine the performance of this technique under varying surveillance settings.

Another study [2], which is an arXiv preprint from 2023, is dedicated to the description of individual humans in surveillance videos. This involves considerations such as appearances, behaviors, locations, and interactions. According to the reviewed literature, there exists a dataset comprising 7,820 individuals in 1,012 videos.

Person-specific captions could offer more detailed data compared to one caption for the whole video clip, but the generation of such captions becomes more complex as the number of individuals increases due to overlapping or partial occlusion of the people in crowded scenarios. The peer-reviewed status of the present research paper also needs to be verified before the submission.

In 2024, the IEEE Systems Journal published an article devoted to the problem of dense video captioning by using multiperspective edge computing with cameras [3]. This method is based on lightweight image stitching, inverted pruned residual architecture, and average sparse attention. Caption quality and computational complexity (in frames per second) are taken into account during evaluation.

However, despite the relevance of the research to the efficient video captioning, the fact that it deals with multiple cameras and edge computing makes the comparison impossible because the performance of this algorithm in ordinary CCTV video streams needs to be tested.

B. Video Captioning for General Purpose

Several research articles have been published in 2024 that focused on enhancing video captioning performance using general purpose benchmarks. Bilingual video captioning [4], which takes advantage of key frame extraction from temporal and visual perspective, alongside LSTM encoder-decoder architectures, was used to provide captioning in English and Arabic. From the results obtained in the research, the English captioning model yielded a BLEU score of 47.18, METEOR score of 30.46, ROUGE-L score of 62.07, and a CIDEr score of 59.98 on MSVD benchmark.

An additional method [5] uses a center-informed framework that enables visual and textual feature alignment across different modalities. This work is tested using the MSVD and MSR-VTT datasets. A dual learning approach [6] that involves caption generation and reconstruction modules has been developed. Furthermore, the authors of this paper note that increasing the number of blocks does not guarantee a better caption but will increase the training difficulty.

An additional research paper [7] has improved the alignment of videos and texts by applying multi-grained alignment and a learnable token-shift Refineformer module.

These researches have provided some good information about video captioning in general. But the datasets used in their experiments belong to the general domain, and thus the results might not be a true reflection of the performance of such systems on CCTV videos due to lower resolution, occlusions, and unpredictable events.

The results mentioned in [4] cannot be directly compared to the ones we achieved in our experiment because of the differences in the datasets and other aspects.

C. Commonsense and Dense Captioning

A transformer based method [8] incorporates knowledge about visuals and language along with commonsense knowledge to come up with sentences that describe the video. The method has been applied on ActivityNet Captions for dense video captioning and ALFRED for instruction generation.

The paper brings out a significant problem: the inclusion of commonsense knowledge may help in generating descriptions that make more sense, however, there is a possibility that the description may contain information that does not appear in the video.This is particularly relevant to surveillance settings.

DCMSTRD [9] proposes an end-to-end solution for dense video captioning via multi-scale transformer decoding and set-based prediction. The architecture minimizes the need for manually engineered modules and simplifies the caption generation process.

Dense video captioning usually involves identification and description of a number of events in a video, which makes it more difficult than creating a caption for an entire video clip. While the methods proposed in these works add value to the overall domain of video understanding, none of them address the task of caption generation for CCTV videos.

These papers prove that video captioning has evolved through different directions starting with surveillance video understanding based on images and ending with dense captioning and vision-language alignment. Nevertheless, it is still possible to explore how a pre-trained vision-language model operates on CCTV-style clips without any fine-tuning. This problem is covered by the current paper in the small-scale experiment with Qwen3-VL-4B-Instruct utilizing uniform frame sampling and caption evaluation metrics.

E. Research Gap

Some topics still require research efforts in video captioning especially in relation to the surveillance video data analysis.

First of all, the majority of currently used approaches for video captioning [4]–[9] were evaluated using standard datasets.

It does not mean that these models are capable of accurately describing scenes from CCTV video data which may include low quality, partial object visibility, and hard-to-analyze activities.

Moreover, another problem relates to the selection of relevant frames from lengthy video clips without analyzing every single frame. Since many adjacent frames have very similar contents, selecting frames periodically can improve the coverage of the model and reduce the complexity of computations [3], [4].

The correctness of the descriptions is another factor. While a model might create fluent descriptions, it might contain data that cannot be observed in the video itself. As noted in [8], the usage of commonsense knowledge may introduce false data. In the case of surveillance, captions should be as true to reality as possible.

Lastly, some systems of surveillance have separate modules for generating the captions and classifying the risks [1]. Thus, it would be interesting to analyze what one pre-trained model of vision-language could produce without an additional classifier or training.

In this work, we analyze the above needs via a small evaluation of Qwen3-VL-4B-Instruct on 10 CCTV and urban-scene videos. We use uniform frame sampling to select the video frames, and we assess the quality of generated captions with common text overlap metrics. We do not claim that this model is superior to those methods described above as we do not conduct any comparative evaluation.

III. Proposed Methodology

The suggested approach generates a text caption for each video using a set of simple operations on videos. The pretrained model Qwen3-VL-4B-Instruct is used without any further training and fine-tuning.

A. Overview and System Flow Chart

Each of the MP4 videos is processed independently. First, the video is loaded, and frames at approximately regular intervals are chosen. Resized frames are fed to the pretrained model, which outputs a caption corresponding to the observed scene. Information about the generated caption and processing procedure is stored, and generated text is compared with the reference caption.

Figure 1 shows the flow chart of the developed system. Only one vision-language model is used to generate captions in the pipeline. There is no object detection model, action recognition model or other caption generator in the pipeline.

The procedure starts with uploading an input video and picking up to eight images from it. Each picked image is scaled such that the maximum side of it is not longer than 768 pixels. After that, these images are fed into Qwen3-VL-4B-Instruct model which produces a single natural language caption for this video.

Caption generation is followed by storing the produced

caption and other processing data in CSV and JSON files.

Finally, BLEU-4, METEOR and ROUGE-L metrics are computed by comparing the produced caption with its reference.

B. Video Input

The system takes as input MP4 video clips with scenes like street scenes, crosswalks, cars, intersections, and buildings. The key idea here is to characterize the scene visually, not trying to guess human intentions or occurrences that are not supported by the video itself.

The database comprises two types of video clips – surveillance video clips and regular city scenes. It is critical since not all video clips come from surveillance cameras.

C. Uniform Sampling of Frames

Sampling each frame of a video may raise the amount of computation required since consecutive frames have similar scenes. In order to minimize this overhead, the algorithm samples eight frames, approximately uniformly distributed over each video clip.

The uniform sampling of the video makes it possible for the algorithm to see different points in time rather than just the starting point of the video. But this method also suffers from one drawback: any activity that occurs between two sampled frames would not get captured.

In the experiment, nine videos had eight frames sampled each and one had seven frames sampled, as noted in Section V.

D. Image Preprocessing

Following frame selection, the images are resized to ensure their maximum side dimension does not exceed 768 pixels. This helps to minimize the volume of visual information sent to the model while maintaining the most important elements of the picture.

Then, the resized pictures are preprocessed using the input-processing interface of the model itself. In other words, no extra image analysis model is needed in this stage.

E. Pretrained Vision-Language Model

The only model utilized for captioning is Qwen3-VL-4B-Instruct. Being a pretrained vision-language model, this network takes the visual input and provides natural language responses.

It is applied directly in the process of inference, meaning that the weights of the model are not trained during the experiment. The absence of any task-specific dataset makes the implementation quite straightforward.

F. Captioning

The preprocessed images are fed to the Qwen3-VL-4B-Instruct model using the vision-language inference API. The model produces one caption that is supposed to convey the contents of the video.

Caption generation is done using the parameters of max_new_tokens set to 180, and do_sample parameter set to False. This ensures the consistency of the output with the same inputs, model, and parameters.

Depending on what the sampled images contain, the generated captions may convey information about the pedestrians, cars, roads, buildings, and other surroundings. Nevertheless, the system does not recognize abnormal events, intentions of people, nor does it give any surveillance warning

G. Output Storage

For each video, information such as the filename, caption generated, video length, frame samples used, generation time, device for inference and inference status is stored.

The outputs are stored in both CSV and JSON formats. These files help in analyzing the captions individually, organizing the experiment results and computing the metrics.

IV. Dataset, Experiment and Evaluation Metric

The dataset for evaluation consists of 10 video snippets that have a total duration of 309.17 seconds. The experiments were all conducted using Google Colab on a Tesla T4 GPU, with each generated caption being evaluated against a single reference caption using BLEU-4, METEOR and ROUGE-L metrics.

A. Description of the Dataset

The dataset comprises of 10 MP4 video files representing various scenes in the urban area like street corners, crossing point for pedestrians, tram lines, junction on the road below a highway, junction at night and pedestrians' walkway.

The length of each video file varies from 7.25 to 119.99 seconds, with an average length of 30.92 seconds. The majority of videos present a fixed, elevated or street-level camera shot. However, in the case of V8, there is drone footage of a city from above; thus, the dataset consists not only of CCTV video clips but also some urban videos.

There is a reference caption that consists of one sentence containing 10 to 18 words for each video. It is used as a reference text for the evaluation process. The description of video files and their resolutions and frame rates are given in Table II.

B. Experimental Environment

All experiments were conducted in Google Colab using a Tesla T4 GPU with CUDA support. The inference hardware associated with all videos is cuda:0.

The Qwen3-VL-4B-Instruct model was adopted with parameters presented in Table III. All 10 videos passed through processing with success status.

C. Evaluation Measures

In order to test the performance of the captions that are generated, three automatic measures were used: BLEU-4, METEOR and ROUGE-L. For each measure, it was computed for each video independently and averaged over the ten videos.

The BLEU-4 measure takes into account how many sequences of up to four consecutive words from the generated caption appear in the reference caption as well as applies a penalty for those captions which are shorter than the reference. Because it heavily depends on matching the word sequences, a caption which means the same thing but differs in wording will still be assigned a poor score.

METEOR measure also checks the words in both the generated and reference captions, but unlike the BLEU-4 measure allows for some differences in wording.

ROUGE-L uses the longest common subsequence of words between the caption and the reference. The sequence has to match in order, but does not need to be contiguous. In this experiment, the ROUGE-L F1 measure is utilized; that is a combination of precision and recall.

While these measures are helpful in assessing text, they alone do not suffice as a means of verification that the caption matches the video. The caption may be visual in content, but still earn a low score because it has used different words than those in the reference. For this reason, the score itself needs to be viewed in conjunction with the caption.

Caption generation time is calculated as well. That gives an idea of speed, but not one of caption quality.

V. Results and Discussion

The generated captions had an average of 0.0101 BLEU-4 score, 0.2342 METEOR score and 0.1339 ROUGE-L score while taking an average caption generation time of 17.40 seconds per video.

A. Caption Quality Scores

Table IV shows the mean score of each performance measure along with the minimum and maximum scores obtained for the videos. Table V shows the individual scores of all the 10 videos, also the length of the generated caption and reference caption in words. The length of the caption was computed using spaces.

B. Discussion of Scores

The length of the captions and metric scores.

The most significant distinction is the length of the captions generated by the model. The generated descriptions have a word count of 107.1 words, whereas the references have 14.0 words. This means that the generated descriptions are seven times longer than the references.

This is important in understanding the scores received because the BLEU-4 metric is based on finding word n-grams that include up to four consecutive words. Longer descriptions will, therefore, likely contain numerous word sequences that do not exist in the short references. The ROUGE-L metric is also impacted because it uses the F1 score that takes into account both the matching words and the length of the description. This partially explains why the BLEU-4 score is quite low and the ROUGE-L score is rather low, even though it does not mean that the descriptions are wrong. They are compared to detailed descriptions and short phrases.

METEOR results. Among the three metrics, the best mean score was reached by METEOR equal to 0.2342. Because it allows for slight variations in word matching, it is less sensitive to the exact phrasing. The video V3 has the highest score among all videos

Best and weakest videos. The best video in terms of BLEU-4 and ROUGE-L scores is V1. The caption contains all the mentioned in the reference elements: winter street, snow, pedestrians walking on the pavement, storefronts and temporary barriers. The generated caption includes many of the same details and has higher textual overlap because of this.

The worst video in terms of BLEU-4 and ROUGE-L scores is V6. The generated caption of this video contains many vehicles and pedestrians, while the reference gives a general description of the scene. Thus, the length and wording of the captions reduce textual overlap.

V8 has the lowest METEOR score among the selected videos. This is explained by the fact that the generated caption describes both a ground-level shot at the beginning and an aerial drone shot in the end of the video, while the reference only describes the second part. Thus, the generated caption refers to more of the video than the reference, but this does not mean that it increases overlap score.

When low scores do not say everything. The next example (V9) shows the situation where low scores require further analysis. This caption talks about night time junction where police cars with blue flashing lights come. However, in the reference there are police lights in the night and BLEU-4 score is only 0.0039.

The example above shows that even if two captions talk about the same thing, they may not have high word-sequence overlap, especially if one caption is much longer than another. Low score should be seen as indication of lack of textual similarity rather than evidence that the caption is wrong.

Details for verification. In some cases, generated captions may contain specific details like readable names of the shops, makes of vehicles and the name of the bridge. However, since each video has only one reference caption, metric scores are unable to confirm or deny these facts. They should be verified by analyzing the original footage.

These results describe the behaviour of one model under one

configuration. Since no other models were used for the task, it is impossible to use scores for ranking Qwen3-VL-4B-Instruct against other methods.

C. Inference-time results

The generation time of a caption was estimated separately from the quality of the caption itself. During testing of the 10 videos, the total generation time recorded was 174.04 seconds, which gives the average generation time of 17.40 seconds per video. The fastest one was V9 – 10.38 seconds, while the slowest one was V4 – 21.28 seconds. Table VI contains generation times of the videos and their caption lengths.

It should be noted that the generation time of a caption does not depend on the duration of a video. The longest video V8, which has the length of 119.99 seconds, takes 21.20 seconds to generate a caption. At the same time, the video V9 lasts only 20.82 seconds but needs 10.38 seconds for processing.

It can be explained by the fact that all the videos are reduced to a maximum of eight resized images, which are used as inputs to the model. Consequently, the video duration does not affect the number of images that will be processed by the model.

Among all tested videos, V4, V8 and V10 were the slowest in generation time and produced the longest captions. V9 has the shortest caption and, consequently, the shortest processing time.

The values above are applicable to the specific hardware and configuration in which the experiment was conducted. It does not show that the system operates in real time.

VI. Limitations

While it is true that the suggested model produces relevant captions to describe the content of a video recorded by CCTV cameras, comprehending complicated actions or interactions of actions is still a difficult issue. There are scenarios where the caption created does not fully depict all the actions within an activity or explain how the event progresses with time. This can be attributed to the performance of the system in terms of video quality, complexity of the scene, or the choice of video frames..

VII. Conclusion and Future Work

A. Conclusion

In this study, a simple video-captioning pipeline based on the pretrained Qwen3-VL-4B-Instruct model was implemented and evaluated. This system samples up to 8 frames from each video at roughly equal intervals, resizes them to have their largest dimensions equal 768 pixels and generates one caption without fine-tuning or further training.

Sampling is switched off, while the maximum number of new tokens is 180.

Evaluation was done for 10 CCTV and urban scene clips. Compared to one reference caption per video, the generated captions had mean BLEU-4, METEOR and ROUGE-L scores of 0.0101, 0.2342 and 0.1339 correspondingly. The average caption generation time was 17.40 seconds per video on a Google Colab Tesla T4 GPU.

Among the main findings is that the generated captions were noticeably longer and provided more information compared to reference captions. This finding explains low text-overlap scores and proves that automatic metrics are not appropriate for judging the quality of video captions.

However, the limitations of this research lie in the use of a very limited dataset, one reference per video, and lack of a baseline. This research does not prove that this approach performs better than others, and the model does not perform event detection or abnormality recognition, it has not been tested in practice as a surveillance system.

B. Future Work

There are several ways how this research can be extended. Firstly, the experiments can be conducted on a larger set of real CCTV recordings taken under various lighting conditions and camera positions and showing different activities. It will allow analyzing how the model operates under different conditions and making more informed conclusions about its capabilities.

Preparation of several references for each video can make the evaluation more fair and compensate for differences in wording and length which affect automatic metrics. Caption accuracy, completeness and unsupported information can also be assessed by humans in addition to automated metrics.

Furthermore, future experiments can investigate the impact of sampling frames quantity and interval on caption generation time and quality. Comparing captions with various frame samples will provide additional information about what kind of moments need to be captured.

Finally, a comparative analysis of Qwen3-VL-4B-Instruct with other captioning models can be conducted on the same set of videos and with the same evaluation settings. In addition, instruction for generating shorter descriptions and evaluation of resulting metrics and readability can be done.

References

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Details to Verify Before Submission

A number of important details need to be sorted out before submitting the paper. These include adding author names and affiliations to the title page as well as verifying the author names, exact titles and publication information for all nine references. The source and license information needs to be provided for each video, especially considering that the videos could potentially contain stock footage or footage from the V8 drones.

The preparation of the captions for references should also be explained in the paper, such as who did the preparation and how the caption was prepared. The exact prompt for Qwen3-VL-4B-Instruct should be provided in section III-F. Information on the model identifier, software versions, metric-library settings and BLEU-4 smoothing parameters would also help.

The method of measuring timing needs to specify whether frame sampling and resampling were also included in the generation time or whether it was just the inference time of the model. Captions containing the names of shops, vehicles and the named bridge also need to be verified using the corresponding videos.

Citations should be added for Qwen3-VL, BLEU, METEOR and ROUGE-L. Fig. 1 and the experimental results need to be verified based on the actual code and data files.